

GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

Machine Parameters (Pre-Defined)

[row 8] GT_FREQ_GHZ : GT Freq (GHz) [row 26] XECU_COUNT : XeCU count [row 27] XECORE_PER_XECU : XeCore per XeCU [row 28] EU_PER_XECORE : EU per XeCore [row 29] L2_BANKS_PER_XECU : L2 banks per XeCU [row 30] BANK_CAPACITY_MB : bank capacity (MB) [row 32] DPAS_DEPTH : DPAS depth [row 35] COMPUTE EFFICIENCY_PCT : compute efficiency % [row 86] L1_READ_MAX_B_EU_CLK : L1 read max (B/EU/clock) [row 87] L1_WRITE_MAX_B_EU_CLK : L1 write max (B/EU/clock) [row 97] GTI_READ_MAX_BW_B_CLK : GTI read max BW (B/clock) [row 98] GTI_WRITE_MAX_BW_B_CLK : GTI write max BW (B/clock) [row 124] MAX_POSSIBLE_HBM_BW_GB_S : max possible HBM BW (GB/s)

Workload Parameters (Pre-Defined)

[row 3] INPUT_A_DATA_FORMAT : input A data format (e.g. fp8) [row 4] INPUT_B_DATA_FORMAT : input B data format (e.g. fp4) [row 5] OUTPUT_D_DATA_FORMAT : output D data format (e.g. fp8) [row 9] M : M dimension [row 10] K : K dimension [row 11] N : N dimension [row 24] OUTPUT_BYTES_PER_ELEMENT_FP32 : output bytes per element (fp32) [row 34] MACHINE_OCCUPANCY_PCT : machine occupancy % [row 39] M_PER_THREAD : M per thread [row 40] K_PER_THREAD : K per thread [row 41] N_PER_THREAD : N per thread [row 45] TG_WIDTH_IN_UNITS_OF_THREAD : TG width in units of thread [row 46] TG_HEIGHT_IN_UNITS_OF_THREAD : TG height in units of thread [row 50] XECORE_CLUSTER_WIDTH_IN_UNITS_OF_TG_KEEP_THE_CLUSTER_SIZE_AS_4 : XeCore cluster width in units of TG (keep cluster size as 4) [row 51] XECORE_CLUSTER_HEIGHT_IN_UNITS_OF_TG_KEEP_THE_CLUSTER_SIZE_AS_4 : XeCore cluster height in units of TG (keep cluster size as 4) [row 54] XECU_TILE_WIDTH_IN_UNITS_OF_TG : XeCU tile width in units of TG [row 55] XECU_TILE_HEIGHT_IN_UNITS_OF_TG : XeCU tile height in units of TG [row 58] GPU_TILE_WIDTH_IN_XECU_UNIT : GPU tile width in XeCU unit

Computed Equations

row 22 - INPUT A BYTES PER ELEMENT

$$\text{Bytes}_{\text{perElement}}^{(\text{matA})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matA})}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\text{Bytes}_{\text{perElement}}^{(\text{matB})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matB})}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\text{Bytes}_{\text{perElement}}^{(\text{matD}\downarrow)} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matD}\downarrow)}]$$

row 33 - EU COUNT

$$|\text{EU}| = |\text{XeCore}|_{\text{perXeCU}} \times |\text{EU}|_{\text{perXeCore}} \times |\text{XeCU}|$$

row 36a - MMA MAC THROUGHPUT PER EU

$$\tau_{mMACp(\text{Clk} \cdot \text{EU})}^{(\text{peak})} = \frac{4}{\max(\text{Bytes}_{\text{perElement}}^{(\text{matA})}, \text{Bytes}_{\text{perElement}}^{(\text{matB})})} \times D_{DPAS} \times 16$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{mMACp(\text{Clk} \cdot \text{XeCore})}^{(\text{peak})} = \tau_{mMACp(\text{Clk} \cdot \text{EU})}^{(\text{peak})} \times |\text{EU}|_{\text{perXeCore}}$$

row 42 - CLKS PER DPAS

$$\text{CLKS}_{DPAS} = \frac{M_{\text{perThread}} \times K_{\text{perThread}} \times N_{\text{perThread}}}{\frac{\tau_{mMACp(\text{Clk} \cdot \text{XeCore})}^{(\text{peak})}}{|\text{EU}|_{\text{perXeCore}}}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$W_{\text{element}}^{(\text{Thread})} = N_{\text{perThread}}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$H_{\text{element}}^{(\text{Thread})} = M_{\text{perThread}}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})} = W_{\text{thread}}^{(\text{ThreadGroup})} \times W_{\text{element}}^{(\text{Thread})}$$

row 14 - TG TILES IN N

$$|\text{Tiles}|_N^{(\text{TG})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$H_{\text{element}}^{(\text{ThreadGroup})} = H_{\text{thread}}^{(\text{ThreadGroup})} \times H_{\text{element}}^{(\text{Thread})}$$

row 15 - TG TILES IN M

$$|\text{Tiles}|_M^{(\text{TG})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{ThreadGroup})}}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$W_{\text{element}}^{(\text{XeCoreCluster})} = W_{\text{TG, keep cluster size as 4}}^{(\text{XeCoreCluster})} \times W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}$$

row 16 - TG CLUSTER TILES IN N

$$|\text{Tiles}|_N^{(TG\ Cluster)} = \frac{N_{dim}}{W_{element}^{(XeCoreCluster)}}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(XeCoreCluster)} = H_{TG, keep\ cluster\ size\ as\ 4}^{(XeCoreCluster)} \times H_{element}^{(ThreadGroup)}$$

row 17 - TG CLUSTER TILES IN M

$$|\text{Tiles}|_M^{(TG\ Cluster)} = \frac{M_{dim}}{H_{element}^{(XeCoreCluster)}}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$W_{element}^{(XeCUTile)} = W_{TG}^{(XECUTile)} \times W_{element}^{(ThreadGroup, realized\ by\ multiple\ MMA\ iterations)}$$

row 18 - XECU TILES IN N

$$|\text{Tiles}|_N^{(XeCU)} = \frac{N_{dim}}{W_{element}^{(XeCUTile)}}$$

row 20 - GPU TILES IN N

$$|\text{Tiles}|_N^{(GPU)} = \frac{\frac{N_{dim}}{W_{element}^{(XeCUTile)}}}{W_{XeCU}^{(GPUTile)}}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(XeCUTile)} = H_{TG}^{(XECUTile)} \times H_{element}^{(ThreadGroup)}$$

row 19 - XECU TILES IN M

$$|\text{Tiles}|_M^{(XeCU)} = \frac{M_{dim}}{H_{element}^{(XeCUTile)}}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$H_{XeCU}^{(GPUTile)} = \frac{|\text{XeCU}|}{W_{XeCU}^{(GPUTile)}}$$

row 21 - GPU TILES IN M

$$|\text{Tiles}|_M^{(GPU)} = \frac{\frac{M_{dim}}{H_{element}^{(XeCUTile)}}}{H_{XeCU}^{(GPUTile)}}$$

row 13 - WAVES

$$N_{waves} = \text{CEILING}(|\text{Tiles}|_N^{(GPU)}, 1) \times \text{CEILING}(|\text{Tiles}|_M^{(GPU)}, 1)$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$W_{element}^{(GPUPTile)} = W_{XeCU}^{(GPUPTile)} \times W_{element}^{(XeCUTile)}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$H_{element}^{(GPUPTile)} = H_{XeCU}^{(GPUPTile)} \times H_{element}^{(XeCUTile)}$$

row 63 - MAT A INPUT SIZE B

$$Size_B^{(matA)} = M_{dim} \times K_{dim} \times Bytes_{perElement}^{(matA)}$$

row 64 - MAT B INPUT SIZE B

$$Size_B^{(matB)} = K_{dim} \times N_{dim} \times Bytes_{perElement}^{(matB)}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$Size_B^{(matC, matD)} = M_{dim} \times N_{dim} \times Bytes_{perElement}^{(matD\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$Size_B^{(matD\downarrow)} = M_{dim} \times N_{dim} \times Byte_{perElement}^{(matD)}$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$L2Size_B = L2BankSize_{MB} \times |L2Banks|_{perXeCU} \times 1024 \times 1024$$

**row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF
THREAD DIVERGENCE**

$$|WorkingSet|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)} = \min(20000 \times \frac{K_{perThread}}{CLKS_{DPAS}}, K_{dim})$$

**row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE
INSTANCE AND SINGLE WAVE**

$$\begin{aligned} TotalRequiredL2Size_{1\ instance,\ 1\ wave}^{matB\ always\ hit} &= H_{element}^{(XeCUTile)} \times Bytes_{perElement}^{(matA)} \times |WorkingSet|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)} \\ &+ W_{element}^{(XeCUTile)} \times Bytes_{perElement}^{(matB)} \times |WorkingSet|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)} \\ &+ W_{element}^{(XeCUTile)} \times H_{element}^{(XeCUTile)} \times Bytes_{perElement}^{(matD\downarrow)} \end{aligned}$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$MemBW_{BpClk}^{max} = \frac{MemBW_{GBps}^{max}}{f_{GHz}^{(GT)}}$$

row 37a - WORKLOAD MAC PER XECORE

$$WL_{MAC}^{(XeCore)} = \frac{M_{dim} \times K_{dim} \times N_{dim}}{|XeCore|_{perXeCU} \times |XeCU|}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$T_{clk}^{(total)} = \frac{WL_{MAC}^{(XeCore)}}{\tau_{mMACp(Cl k \cdot XeCore)}^{(peak)} \times \eta_{systolic}}$$

row 69 - TOTAL L2 READ B

$$\begin{aligned} L2Rd_B^{(total)} &= Size_B^{(matA)} \times CEILING(\frac{N_{dim}}{W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}}, 1) \\ &+ Size_B^{(matB)} \times CEILING(\frac{M_{dim}}{H_{element}^{(ThreadGroup)}}, 1) \end{aligned}$$

row 70 - TOTAL L2 WRITE B

$$L2Wr_B^{(total)} = Size_B^{(matC, matD)}$$

row 71 - TOTAL L1 READ B

$$\begin{aligned} L1Rd_B^{(total)} &= Size_B^{(matA)} \times CEILING(\frac{N_{dim}}{W_{element}^{(Thread)}}, 1) \\ &+ Size_B^{(matB)} \times CEILING(\frac{M_{dim}}{H_{element}^{(Thread)}}, 1) \end{aligned}$$

row 72 - TOTAL L1 WRITE B

$$L1Wr_B^{(total)} = Size_B^{(matC, matD)}$$

row 74 - L2 READ B XECORE CLK

$$L2RdBW_{Bp(Cl k \cdot XeCore)} = \frac{\frac{L2Rd_B^{(total)}}{|XeCU| \times |XeCore|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 75 - L2 WRITE B XECORE CLK

$$L2WrBW_{Bp(Cl k \cdot XeCore)} = \frac{\frac{L2Wr_B^{(total)}}{|XeCU| \times |XeCore|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\begin{aligned} L2RdWrBW_{Bp(Cl k \cdot XeCore)} &= L2RdBW_{Bp(Cl k \cdot XeCore)} \\ &+ L2WrBW_{Bp(Cl k \cdot XeCore)} \end{aligned}$$

row 79 - L1 READ B EU CLK

$$L1RdBW_{Bp(Clk \cdot EU)} = \frac{\frac{L1Rd_B^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

row 77 - L1 READ B XECORE CLK

$$L1RdBW_{Bp(Clk \cdot XeCore)} = L1RdBW_{Bp(Clk \cdot EU)} \times |EU|_{perXeCore}$$

row 80 - L1 WRITE B EU CLK

$$L1WrBW_{Bp(Clk \cdot EU)} = \frac{\frac{L1Wr_B^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

row 78 - L1 WRITE B XECORECLK

$$L1WrBW_{Bp(Clk \cdot XeCore)} = L1WrBW_{Bp(Clk \cdot EU)} \times |EU|_{perXeCore}$$

row 83 - L2 READ MAX B XECORE CLK

$$L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)} = \frac{|XeCore|_{perXeCU} \times 64}{|XeCore|_{perXeCU}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{L2RdBWpXeCore} = \frac{L2RdBW_{Bp(Clk \cdot XeCore)}}{L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{L2WrBWpXeCore} = \frac{L2WrBW_{Bp(Clk \cdot XeCore)}}{L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{L2RdWrBWpXeCore} = \frac{L2RdWrBW_{Bp(Clk \cdot XeCore)}}{L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{L1RdBWpEU} = \frac{L1RdBW_{Bp(Clk \cdot EU)}}{L1RdBW_{Bp(Clk \cdot EU)}^{max}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{L1WrBWpEU} = \frac{L1WrBW_{Bp(Clk \cdot EU)}}{L1WrBW_{Bp(Clk \cdot EU)}^{max}}$$

**row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING
DATA SET PCT**

$$L2HitRate_{random \ WS \ access} = \min\left(\frac{L2Size_B}{TotalRequiredL2Size_{matB \ alwayshit}^{1 \ instance, \ 1 \ wave}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$L2MissRate = 1 - L2HitRate_{random \ WS \ access}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$L2RdB_{total} = L2RdB_B^{(total)}$$

**row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE
PCT**

$$\begin{aligned} P_{L2Hit, after \ 1st \ wave}^{(matA)} &= IF(Size_B^{(matA)} \\ &+ Size_B^{(matB)} \\ &+ Size_B^{(matC, matD)} \leq L2Size_B, 1.0, IF(K_{dim} > 2 \times |WorkingSet|_{20K \ clks \ of \ thread \ divergence}^{(K \ in \ L2)}, 0.0, 1 \\ &- \frac{K_{dim} - |WorkingSet|_{20K \ clks \ of \ thread \ divergence}^{(K \ in \ L2)}}{|WorkingSet|_{20K \ clks \ of \ thread \ divergence}^{(K \ in \ L2)}})) \end{aligned}$$

**row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE
PCT**

$$\begin{aligned} P_{L2Hit, after \ 1st \ wave}^{(matB)} &= IF(Size_B^{(matA)} \\ &+ Size_B^{(matB)} \\ &+ Size_B^{(matC, matD)} \leq L2Size_B, 1.0, 0.0) \end{aligned}$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE

PCT

$$P_{L2Miss, after 1st wave}^{(matA)} = 1 - P_{L2Hit, after 1st wave}^{(matA)}$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE

PCT

$$P_{L2Miss, after 1st wave}^{(matB)} = 1 - P_{L2Hit, after 1st wave}^{(matB)}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER

COLD CACHE

$$\begin{aligned} \text{MemRdB}_{cold start}^{(wave)} &= \frac{\left(\text{MemRdB}_{cold start}^{(wave)} \right)_{\text{num}}}{2} \\ \left(\text{MemRdB}_{cold start}^{(wave)} \right)_{\text{num}} &= \text{Size}_B^{(matA)} \\ &+ \text{Size}_B^{(matA)} \times \left(\text{CEILING} \left(\frac{N_{dim}}{W_{element}^{(XeCUTile)}}, 1 \right) - 1 \right) \times P_{L2Miss, after 1st wave}^{(matA)} \\ &+ \text{Size}_B^{(matB)} \\ &+ \text{Size}_B^{(matB)} \times \left(\text{CEILING} \left(\frac{M_{dim}}{H_{element}^{(XeCUTile)}}, 1 \right) - 1 \right) \times P_{L2Miss, after 1st wave}^{(matB)} \\ &+ \left(\left(\text{MemRdB}_{cold start}^{(wave)} \right)_{\text{num}} \right)_{\text{aux4}} \times \text{L2MissRate} \\ \left(\left(\text{MemRdB}_{cold start}^{(wave)} \right)_{\text{num}} \right)_{\text{aux4}} &= \text{L2RdB}_{total} \\ &- \left(\left(\left(\text{MemRdB}_{cold start}^{(wave)} \right)_{\text{num}} \right)_{\text{aux4}} \right)_{\text{aux1}} \end{aligned}$$

$$\begin{aligned}
& \left(\left(\left(\text{MemRdB}_{cold\ start}^{(wave)} \right)_{num} \right)_{aux4} \right)_{aux1} = \text{Size}_B^{(matA)} \\
& + \text{Size}_B^{(matA)} \times \left(\text{CEILING} \left(\frac{N_{dim}}{W_{element}^{(XeCUTile)}}, 1 \right) - 1 \right) \times P_{L2Miss, after\ 1st\ wave}^{(matA)} \\
& + \text{Size}_B^{(matB)} \\
& + \text{Size}_B^{(matB)} \times \left(\text{CEILING} \left(\frac{M_{dim}}{H_{element}^{(XeCUTile)}}, 1 \right) - 1 \right) \times P_{L2Miss, after\ 1st\ wave}^{(matB)}
\end{aligned}$$

row 117 - TOTAL HBM WRITE B

$$\text{MemWrB}^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 118 - TOTAL HBM B

$$\begin{aligned}
\text{MemRdWrB}^{(total)} &= \text{MemWrB}^{(total)} \\
&+ \text{MemRdB}_{cold\ start}^{(wave)}
\end{aligned}$$

row 121 - HBM READ B CLK

$$\text{MemRdBW}_{BpClk} = \frac{\text{MemRdB}_{cold\ start}^{(wave)}}{T_{clk}^{(total)}}$$

row 122 - HBM WRITE B CLK

$$\text{MemWrBW}_{BpClk} = \frac{\text{MemWrB}^{(total)}}{T_{clk}^{(total)}}$$

row 123 - HBM TOTAL B CLK

$$\text{MemRdWrBW}_{BpClk} = \frac{\text{MemRdWrB}^{(total)}}{T_{clk}^{(total)}}$$

row 126 - HBM BW PCT

$$\eta_{MemBW} = \frac{\text{MemRdWrBW}_{BpClk}}{\text{MemBW}_{BpClk}^{max}}$$

row 127 - GTI READ BW PCT

$$\eta_{GTIRdBW} = \frac{\text{MemRdBW}_{BpClk}}{\text{GtiRdBW}_{BpClk}^{max}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{GTIW_{rBW}} = \frac{\text{MemWrBW}_{BpClk}}{\text{GtiWrBW}_{BpClk}^{max}}$$